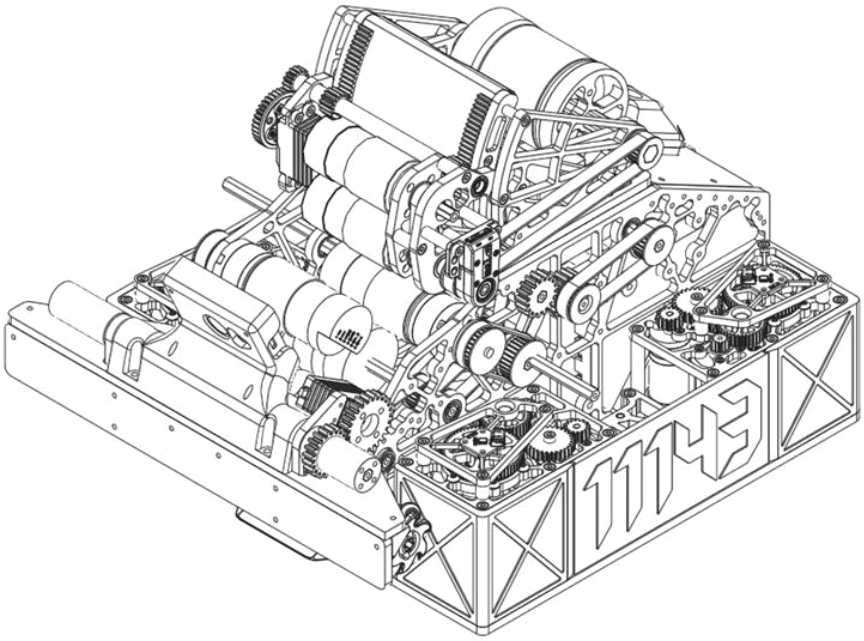
ENGINEERING PORTFOLIO

FTC - FIRST TECH CHALLENGE

BARKER YELLOWBACKS

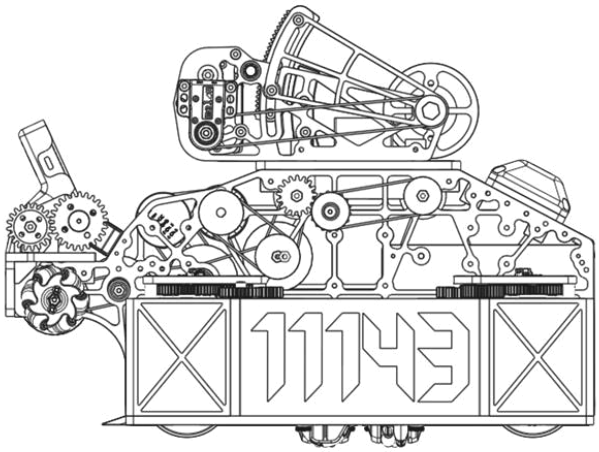
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2026-2027

Sydney
Australia

Barker College
Supported by Barker Robotics



MEET THE TEAM

13 MEMBERS

EB — **Ethan B.**
LEAD DESIGN
assembly

BL — **Brandon L.**
CAD
assembly

IH — **Isaac H.**
LEAD CODE
CAD, assembly

RT — **Ryan T.**
DRIVER
CAD

WQ — **Will Q.**
LEAD STRATEGY
drive coach

IW — **Ivy W.**
PROGRAMMING
HR

TX — **Tina X.**
CODE
assembly

CW — **Chelsea W.**
CODE
assembly

CL — **Chloe L.**
ELECTRICAL
assembly

CS — **Chloe S.**
STRATEGY
code, assembly

CC — **Cooper C.**
SCOUTING
assembly

JC — **Jemima C.**
PROGRAMMING
scouting

LT — **Lucas T.**
CAD
assembly

HOW WE SPLIT THE WORK

- **CAD:** 4 members in Onshape, with our lead designer
- **Code:** 6 members write and test the code
- **Strategy:** 2 strategists plus 2 scouts
- **Assembly:** 9 of 13 members build

HOW WE WORK

- We meet **every week or two**
- Each specialty decides its own area (mechanics decide mechanical changes), then tells the whole team
- Each subsystem has its own code branch, merged into one robot

OUR SEASON

2026 SO FAR (DATES FROM OUR CODE HISTORY)

23 SEP	12 OCT	20 OCT	25 OCT	12 NOV
Shooter	**No turret**	**Intake**	**One robot**	**Wollongong**
first shooter code	turret removed	intake and transfer code	all subsystems merged	first event

WHAT WE KEPT FROM LAST SEASON

Last season's robot (background only) taught us:

- **Swerve works for us:** we kept it
- **A turret takes months to tune:** April to June 2026, so this year we dropped it
- Logging and replay carried over to this season's code

THIS SEASON'S GOALS

- Shoot from **anywhere** on the field
- **Less waiting** between shots
- Collect along the wall and shoot **without turning round**
- Hold our spot against **defence**
- Keep the robot **light**, so it moves more efficiently

Where we are now

The robot is designed in CAD and its code runs in our simulator. **It is not built yet**, so nothing has been tested on a real robot.

TEACHING DARKINJUNG STUDENTS

WHAT

7 members of our team teach robotics to Darkinjung students **once a fortnight**, since about February 2026.

WHAT THEY LEARN

They built robots for this year's **VEX IQ** game and learnt how to build, how to code, how to do interviews and how to write portfolios.

IMPACT

Some of the students were in competitive teams and won **Skills**, **Teamwork Champion** and **Excellence** awards.

ROUND SQUARE INTERNATIONAL CONFERENCE

WHAT

On **28 September 2026** about **7** of us ran the robotics part of the conference at Barker.

REACH

We showed our robotics program to **over 1,500 people** from around **32 countries**, and introduced them to FIRST.

MENTORING VEX TEAMS

Ethan B. and Isaac H. mentor VEX teams.

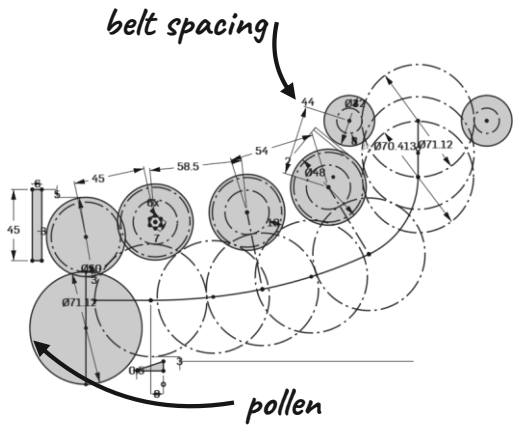
BARKER VEX COMPETITION

Ethan B., Isaac H. and Brandon L. volunteered at the Barker VEX competition.

OUR DESIGN LOOP

1	2	3	4	5
Goals	**Sketch + CAD**	**Simulate**	**Build**	**Test + change**
what the game needs	Onshape layouts	same code, desktop sim	print, cut, assemble	back to step 2

EXAMPLE: FROM SKETCH TO PART



Layout sketch first: pollen size, roller sizes and belt spacing (mm)

ONE ROBOT, MANY BRANCHES

Each subsystem was coded on its own branch: **intake** (Tina X.), **shooter** (Jemima C.), then merged into one robot on **25 October**.

What merging taught us

- Two branches each hid the other's subsystem: fixed
- The shot table divided by zero at its first point: fixed and tested
- Units were mixed (RPM, ticks, degrees): one unit per value now

IDEAS WE BROUGHT IN

FROM VEX

Ethan B., Will Q., Ryan T. and Tina X. played VEX **Push Back**. The game was very similar, so our matchloader follows the same idea (page 10).

FROM LAST SEASON

Swerve and logging carried over; the turret did not (page 9).

HOW WE PLAY

<our scoring plan: what we score first, where we shoot from, our role with the alliance partner>
Pollen collects along the wall, so we collect there and shoot from there.

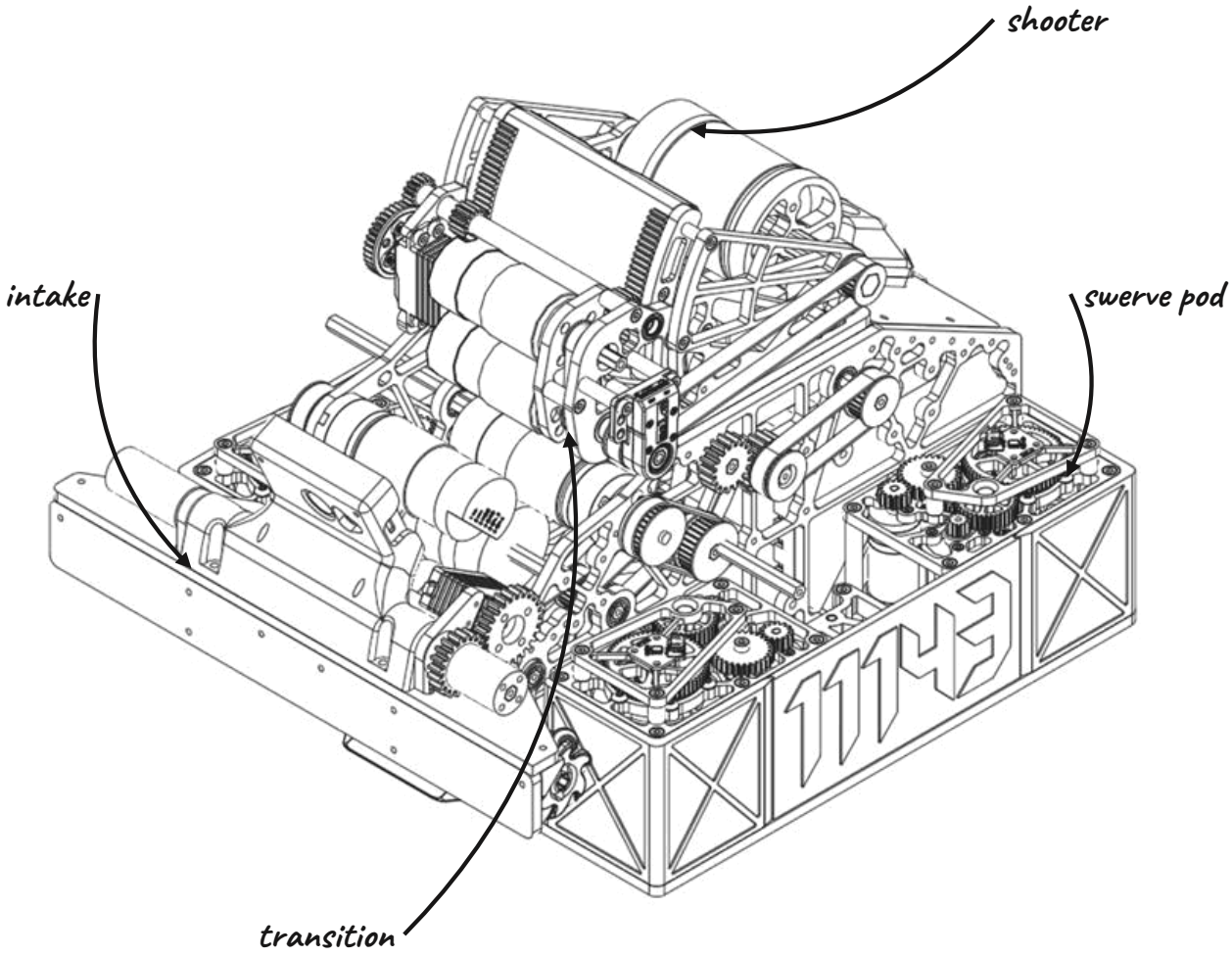
REQUIREMENTS, AND HOW THE ROBOT MEETS EACH

WE NEED TO	WHY	OUR ANSWER
Shoot from anywhere	fewer trips across the field	moving hood + shot table
Shoot quickly, again and again	less waiting between shots	two flywheels
Shoot without turning round	pollen collects along the wall	rear-facing shooter
Aim while driving	save time every cycle	swerve + auto-aim
Hold our spot	defence will push us	swerve X-lock
Load the flower fast	<why>	matchloader
Find pollen and the goal	drive and aim without guessing	front + rear cameras

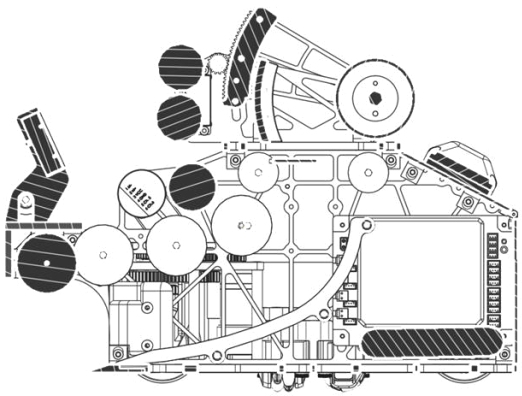
SCOUTING

Cooper C. and Jemima C. scout: <what we record and how it changes our plan>

OUR ROBOT



POLLEN PATH



In at the front, up the transition, out of the rear shooter

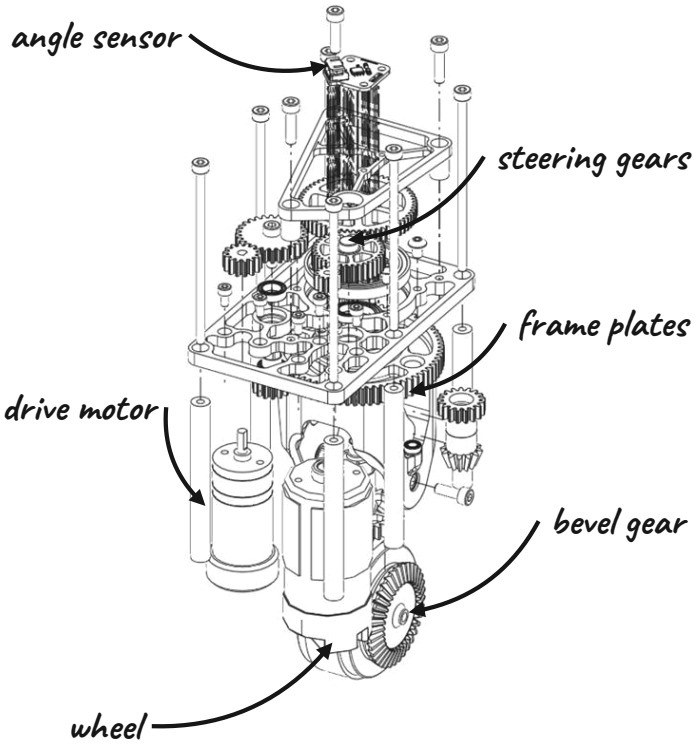
NUMBERS

ITEM	VALUE
Length (CAD)	407.7 mm
Width (CAD)	327.0 mm
Height (CAD)	290.9 mm
Motors	8
Servos	8
Cameras	2

SWERVE DRIVETRAIN

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ONE SWERVE POD



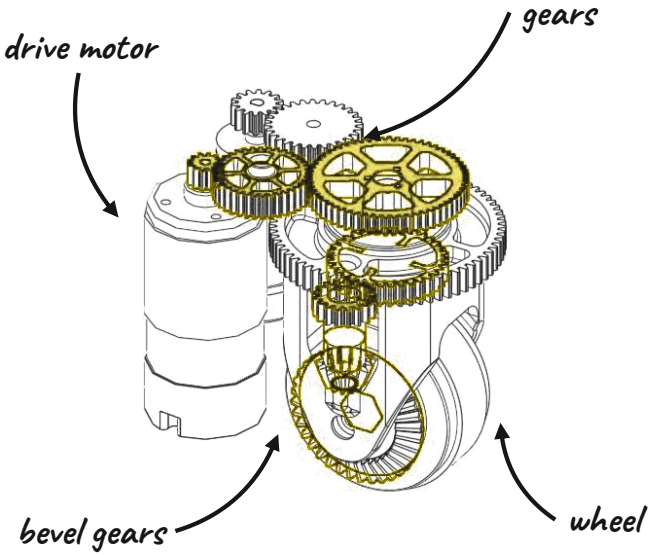
WHY SWERVE, AGAIN

- **X-lock:** the wheels sit in an X, and the friction makes us a lot harder to push or defend
- Moves a lot more **efficiently**: a big advantage in matches
- Full-grip wheels: more **speed and pushing power** than mecanum
- **Aim while driving:** drive one way, face the goal
- We built swerve last season, so the risk was low

Key features

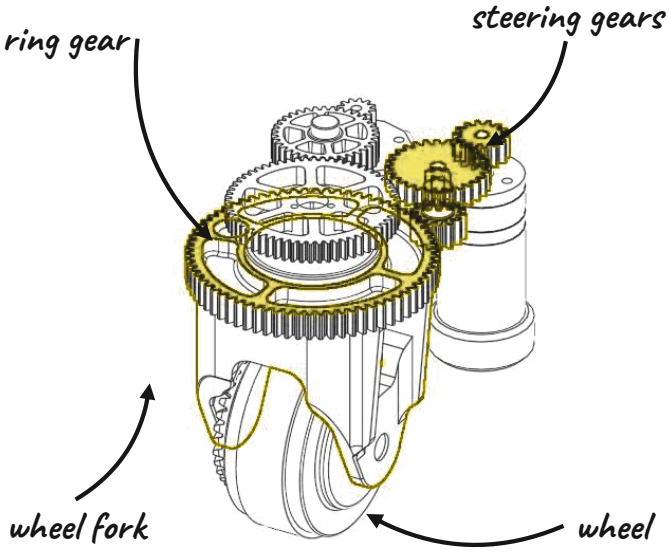
- **4 pods**, each with a drive motor and a steering servo
- An **absolute angle sensor** in every pod, so each wheel knows where it points from the start
- Pod centres **239 x 239 mm** apart; wheels **60.9 mm**

WHEEL DRIVE



Motor, gears, then a bevel gear turn the wheel

MODULE ROTATION



Servo and ring gear turn the whole pod

TESTS

TEST	RESULT	STATUS
Top speed	not built yet	**NOT TESTED**
Pushed by another robot in X-lock	not built yet	**NOT TESTED**

INTAKE AND TRANSITION

WHAT WE TRIED

V1 PRINTED FUNNELS

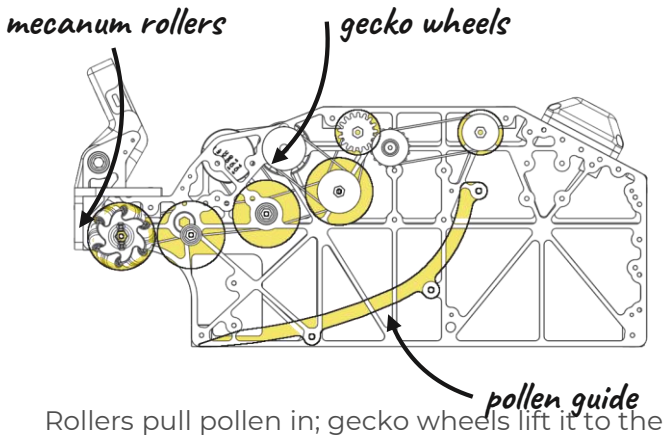
+ <what worked>
− gecko wheels at the front only
− <what went wrong>

→

V2 ROLLERS + TRANSITION

+ mecanum rollers sweep pollen in
+ gecko wheels grip it and carry it up
− <downside, if any>

POLLEN PATH



Rollers pull pollen in; gecko wheels lift it to the shooter

KEY SIZES

- Pollen **71.12 mm**
- Intake mecanum rollers **50 mm**
- Gecko wheels **48 mm**
- Small gecko wheels **32 mm** at the top, so less **backspin** goes into the shooter
- Belt spacing **45, 58.5, 54, 44 mm**
- 2 motors drive the intake and transition
- **Polycarbonate ramps** give the pollen a low-friction surface

MATHS: WHERE THE POLLEN TOUCHES

Our two rows of gecko wheels sit apart, so each row meets the pollen off its centre. The sketch plane cuts the pollen in a circle of **70.413 mm**:

$d = \sqrt{(35.56^2 - 35.21^2)} = 5.0$ mm

So the plane is 5.0 mm from the pollen's centre, and the sketch uses 70.413 mm there, not 71.12 mm.

SENSING

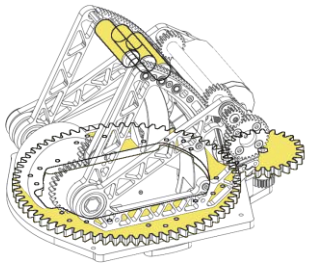
- **3 beam sensors** (bottom, middle, top) count the pollen we hold
- A **gate servo** holds pollen until the shooter is ready
- Jam rate: not tested yet (robot not built)

WHAT WE TRIED

TURRET, LAST SEASON

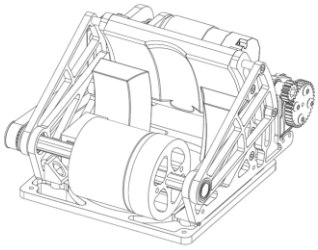
+ aims without turning the robot
− months of tuning (Apr-Jun 2026)

TURRET, THIS SEASON



+ what we wanted first
− weight and space
− 1 more motor or servo
− no time to build it

FIXED REAR SHOOTER



+ fewer parts
+ shoots out of the flowers without turning
− the whole robot must turn to aim

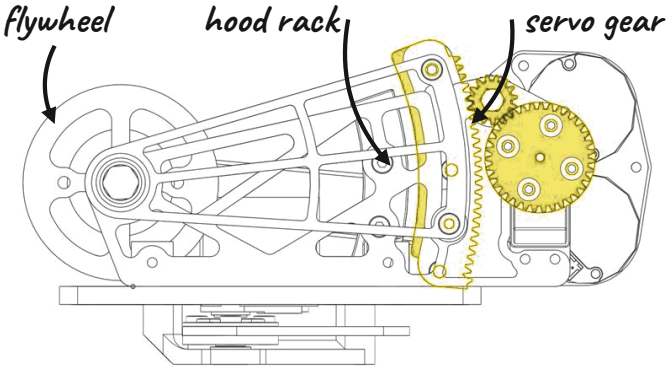
WHAT WE WANTED, AND WHY

- **Two flywheels:** more flywheel means **less waiting** between shots
- **Moving hood:** shoot from anywhere on the field
- **Facing backwards:** pollen collects along the wall, so we drive up to the wall and aim, instead of collecting and then turning round
- **Lookup table:** the code picks flywheel speed and hood angle from where localisation says we are (values still to be measured)

What we changed

The turret got as far as CAD and code, never built. On **12 October** we removed it: the swerve base turns the robot to aim instead (page 13).

HOOD DRIVE



A servo gear drives a rack to tilt the hood

TESTS

TEST	RESULT	STATUS
Shots in from 3 distances	not built yet	**NOT TESTED**
Time between shots	not built yet	**NOT TESTED**

MATCHLOADER AND CAMERAS

MATCHLOADER

IDEA

Four of us played VEX **Push Back**, a very similar game. We used the same kind of matchloader for the flower.

PROBLEM

<what the matchloader fixes, with a number>

RESULT

Not tested yet: the robot is not built.

BLOCKER AND GATE

BLOCKER

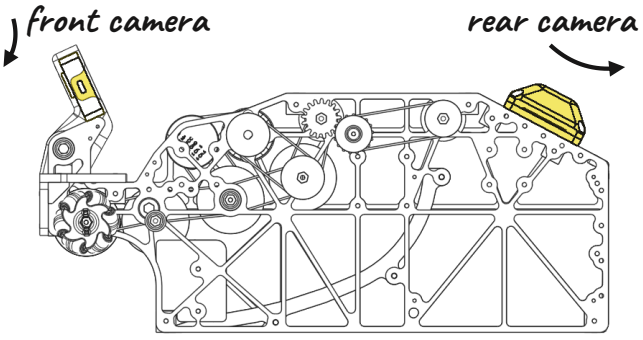
<what the blocker stops> (1 Axon Max servo)

GATE

Stops pollen going into the shooter until the shot is ready (1 Axon servo).

TWO CAMERAS

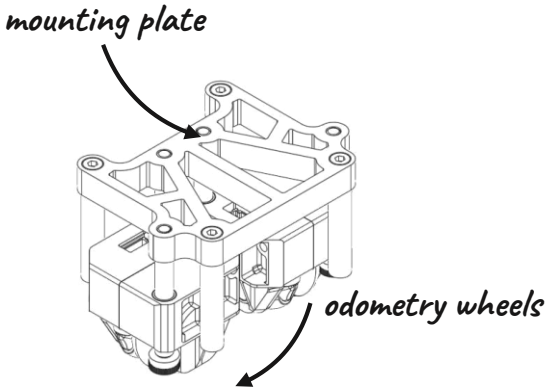
- **Front camera:** finds pollen, so we can path to it
- **Rear camera:** reads AprilTags, so we know where the goal is
- Status: not tested yet (robot not built)



WHAT EACH PART DOES

PART	PROBLEM IT SOLVES	WHAT IT DOES
4 drive motors	moving in any direction	drive each swerve wheel
4 steering servos + 4 angle sensors	pointing every wheel the right way	turn each pod and read its angle
IMU	knowing which way we face	heading for driving and aiming
Front and rear cameras	finding pollen and the goal	pollen pathing; AprilTags
3 beam sensors	knowing how much pollen we hold	sense pollen at bottom, middle and top
Gate servo	pollen reaching the shooter too early	holds pollen until the shot is ready
2 shooter motors	the same shot every time	flywheels held at target speed (PIDF)
Hood servo	shooting from different distances	sets the launch angle from our shot table
Voltage and current checks	brown-outs and stalled motors	logged every loop

ODOMETRY POD



Designed in CAD; the code uses wheels + IMU for now

SPEED CONTROL

CHALLENGE

Each shot takes energy from the flywheels, and a slower wheel shoots short.

SOLUTION

A **PIDF** controller holds the target speed; feedforward uses battery voltage, so a flat battery does not change the shot.

RESULT

Not tested yet: the robot is not built.

SAME CODE ON THE ROBOT AND ON A LAPTOP

PART	WHAT IT HOLDS	WHY
RobotCode	behaviour: driving, aiming, shooting	written once, used everywhere
TeamCode	the real motors, servos and sensors	only place that touches hardware
RobotSim	simulated robot, field and pollen	test before the robot is built

WHY WE SIMULATE

CHALLENGE

The robot is not built yet, but the code needs testing now.

SOLUTION

The same TeleOp runs in our simulator on the official field CAD.

RESULT

Simulation: scores **3 of 3** for both alliances, also while moving.

AUTOMATED TESTS

- **24 desktop tests** pass (25 October 2026)
- They cover heading wrap, 10 ms loops, a lost heading sensor, manual turn override and stop outputs
- Every loop is logged, so a run can be replayed

RISKS AND HOW WE REDUCE THEM

RISK	HOW WE REDUCE IT
Simulation is not the real robot	calibrate hood travel, release point and wheel speed once the robot is built, before shooting for real
Heading sensor lost mid-match	tested in simulation: the robot stops driving and stops feeding pollen
Flat battery changes the shot	feedforward scaled by battery voltage
A complex turret fails	no turret: the drivetrain aims

AUTO-AIM

THE PROBLEM

CHALLENGE

We dropped the turret, so the shooter faces backwards and cannot turn on its own.

SOLUTION

Holding the aim button turns the **whole robot** until it is within **2.5 degrees** of the goal; the driver can still drive.

MATHS: TURNING TO AIM

Turn speed = **feedback** on the heading error + **damping** + **feedforward** for a goal that moves as we drive + a small push past **static friction**.

Gains in our code: kP 4.0, kD 0.35, kS 0.06.

In simulation aiming first stopped **12 degrees** short: small commands were inside the static friction dead zone. Adding voltage feedforward fixed it.

MATHS: SHOOTING WHILE MOVING

The pollen leaves with the shooter's speed **plus the robot's**:

v pollen = v shot + v robot + turn rate × r

(r = where the pollen leaves, from the robot's centre). The solver aims so the sum hits the goal, and the shot table picks flywheel speed and hood angle from the distance.

TESTS

TEST	SIMULATION	ROBOT
Score while still	3 of 3	not built yet
Score while moving	3 of 3	not built yet

OUR ROUTE

<auto route drawing: the team's hand-drawn route, redrawn cleanly>

EACH STEP

STEP	WHAT	POINTS
1	<step>	<pts>
2	<step>	<pts>
3	<step>	<pts>

STATUS

- The autonomous for this robot is **not written yet**: NOT TESTED
- It will use the rear camera's AprilTags to know where we are
- <date we plan to have it running>

TESTING AND LESSONS

TEST RESULTS

TEST	RESULT	STATUS
24 automated code tests	all pass	**PASS**
Simulation: score 3 of 3, still and moving	3 of 3	**PASS**
Robot: shots in from 3 distances	not built yet	**NOT TESTED**
Robot: top speed	not built yet	**NOT TESTED**
Robot: jams per 20 pickups	not built yet	**NOT TESTED**
Autonomous	not written yet	**NOT TESTED**

LESSONS WE APPLIED

- A turret cost months last season, so we let the drivetrain aim
- Sketch the pollen path before CAD: sizes and spacing first
- One unit for each value in code (RPM, radians)

NEXT

- Calibrate the shooter on the real robot
- Write and test the autonomous
- Build the robot, then run every test on this page
- <next design change>

RELIABILITY

No matches yet: the robot is not built. For every match we will record whether it worked all match, and any fault and its fix.

Endnote: text drafted and laid out with help from an AI assistant. All results, photos, CAD, code and decisions are the team's own; the team checked every page.